

Interview Questions and Answers

Q1: Can you explain your experience with developing the Diabetes Disease Prediction System and how you ensured the model's accuracy?

A1: In the Diabetes Disease Prediction System, I developed an end-to-end model using Scikit-learn and XGBoost. I ensured the model's accuracy by performing preprocessing, feature scaling, and thorough model evaluation. The model achieved 87% prediction accuracy. I built a Flask-based web application with REST API integration for instant predictions, which was tested using Postman and deployed on Render.

Q2: How do you integrate LLM APIs into your Streamlit dashboards, and can you give an example of a project where you did this?

A2: I integrate LLM APIs into my Streamlit dashboards by creating an interactive interface where users can input data or prompts, and the dashboard uses the LLM API to generate intelligent predictions or insights. An example of this is the interactive Streamlit dashboard I developed for the Intelligent Systems Workshop at IIT Palakkad, where I integrated LLM API for intelligent predictions and data-driven insights.

Q3: What specific tools and libraries did you use in your Smart Waste Management System, and how did you implement real-time computer vision?

A3: In the Smart Waste Management System, I used Python, YOLOv8 for real-time computer vision, OpenCV for image processing, and Jetson Nano for edge AI. I implemented real-time computer vision by training the YOLOv8 model to detect different types of waste and using OpenCV to process live camera feeds. The system was designed to classify waste in real-time and provide insights for smart waste management.

Q4: Can you describe your experience with deploying machine learning models in a production environment, and what tools did you use?

A4: I have experience deploying machine learning models in a production environment, such as the Diabetes Disease Prediction System. I used Flask to build the web application and deployed it on Render. For the Smart Waste Management System, I used Jetson Nano for edge deployment and integrated the model with sensors and GPIO for real-time processing. I also used Postman for API testing and ensured the models were optimized for performance and scalability.

Q5: How do you handle feature engineering in your projects, and can you give an example from your diabetes prediction system?

A5: In my projects, I focus on thorough feature engineering to improve model performance. For the diabetes prediction system, I performed feature scaling and selection to ensure that the features were appropriately normalized and relevant. I used techniques like correlation analysis and domain knowledge to select the most predictive features. This process helped in achieving the 87% prediction accuracy of the model.